Christopher Francisque

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EDUCATION

University of Pennsylvania — Philadelphia, PA
Master of Computer Science — June 2027

University of Massachusetts Amherst — Amherst, MA
B.S. Mechanical Engineering — June 2024

SKILLS

Languages: Python, C++, JavaScript/Typescript, MATLAB, Java, SQL, Bash
AI/ML: PyTorch, Transformers, Hugging Face, LLM Fine-tuning, LoRA, NumPy, Pandas
Tools: Google Cloud Platform, AWS, Git, Linux, Weights & Biases, Jupyter Notebook, Docker

EXPERIENCE

General Dynamics Mission Systems — Dedham, MA
Software Engineer — Jan 2026 – Present

- Built automated pipeline to parse C++ source and generate unit test plans leveraging the OpenAI API, eliminating ~80% of manual test-planning effort and saving 100+ engineering hours per month.
  Built a Python-based key generation script to test device handling of multiple encryption key formats, achieving 100% accuracy across 264 test vectors and verifying expected behavior for edge cases
- Created a GitLab CI/CD automation script that eliminated build failures on new runners

Brandeis University — Waltham, MA
Machine Learning Research Assistant — June 2025 – Jan 2026

- Implemented methods from 3 research papers, achieving 82-90% accuracy, while only training 0.037% of parameters
- Developed pruning pipelines using learned binary masks to identify redundant parameters, improving weak models by 4%
- Created a gradient tracking system to analyze parameter evolution during training, identified critical embedding layers whose removal caused complete model collapse despite being less than 0.01% of weights
- Built distributed training pipeline across 8 TPU cores handling 110M parameters and 67K samples, solved PyTorch XLA synchronization deadlocks that were causing training failure

Arcada — San Francisco, CA
Software Engineering — June 2025 – Aug 2025

- Identified security vulnerability by implementing Cloudflare infrastructure and rate limiting strategy, reducing bot attacks by 80%
- Scaled user growth to 100K+ unique users and generated 112K in revenue in 3 months from customers such as OpenAI and Google
- Owned end-to-end feature development for core platform capabilities including voting algorithm design and scroll pagination.
- Improved product performance by 30% through data-driven optimization initiatives; identified bottlenecks via user behavior analysis, defined requirements for database query improvements, and validated impact through A/B testing

Raytheon — Andover, MA
Manufacturing Engineer — June 2022 – June 2025

- Led many manufacturing improvement projects spanning root cause analysis, automation programming, fixture design, and training standardization, delivering $160,000+ in cost savings, 10–30% cycle-time reductions, and a 25% cut in operator certification time
- Developed Python and C++ scripts to automate production data analysis for defect rates, operator performance, and statistical process control

PROJECTS

Self-Driving Toy Car — Jan 2026 – Present

- Designed and integrated autonomous RC car using Raspberry Pi 5 and PCA9685 for precise PWM control of motors and servos
- Trained a convolutional neural network using PyTorch to map raw camera frames to continuous steering and throttle outputs, deploying the model for real-time inference on edge hardware
- Integrated camera capture, CNN inference, and servo actuation into a closed-loop control system at 15+ Hz with sub-220ms end-to-end latency

Aircraft 6 Degree-of-Freedom Simulation | Code — December 2024 – March 2025

- Developed a high-fidelity simulation in C++, using physics-based model and implementing numerical integration for accurate flight dynamics
- Optimized computational performance by leveraging efficient data structures to minimize memory overhead, refining numerical integration for stability, and reducing redundant calculations in aerodynamic and atmospheric models to enhance real-time simulation accuracy
- Integrated data visualization using matplotlibcpp to analyze simulation results and validate against NASA data